

VL 326: Fundamentals of Processor Architecture and SoC Design

Course Code: VL 326

Course Name: Fundamentals of Processor Architecture and SoC Design

L-T-P-C: 3-0-2-4

Prerequisites: Digital System Design (EE 210) or equivalent

Intended for: Discipline Core for BTech in VLSI & Microelectronics Engineering Students

Approval: 61st BoA

Preamble

This course provides a hardware-centric introduction to processor architecture and System-on-Chip (SoC) design, bridging classical computer organization concepts with contemporary processor and SoC implementation methodologies. The course begins with foundational computer organization principles—datapath, control, memory, interrupts, and DMA—and progressively advances toward modern pipelined processors, interconnects, and SoC integration using RTL-based design approaches. Emphasis is placed on open-source RISC-V processors, bus-based communication, and hardware/software interaction, supported by hands-on RTL labs and SoC-level projects.

Course Contents

- **Unit-1: Foundations of Computer Organization:** Digital system hierarchy; Von Neumann vs Harvard architectures; Instruction execution cycle (fetch–decode–execute–writeback); Introduction to ISA vs microarchitecture; RISC-V ISA overview (instruction formats, addressing modes); Mapping instruction flow to hardware blocks. (5 hours)
- **Unit-2: Processor Datapath and Control:** Single-cycle processor organization; Multi-cycle datapath design; FSM-based control unit; Microprogrammed control concepts; RTL modeling of datapath and controller. (7 hours)
- **Unit-3: Pipelined Processor Architecture:** Pipeline stages and timing; Structural, data, and control hazards; Hardware techniques: Forwarding, Stalling; Introduction to performance metrics (CPI, throughput). (6 hours)
- **Unit-4: Memory System Organization:** Memory hierarchy fundamentals; Cache organization: direct-mapped, set-associative; Write policies and replacement strategies; Memory-mapped I/O; Introduction to virtual memory (conceptual level). (6 hours)
- **Unit-5: I/O, Interrupts, and DMA:** Programmed I/O vs interrupt-driven I/O; Interrupt handling mechanisms; DMA architecture and operation; DMA controller hardware overview; Role of DMA in SoC performance. (4 hours)
- **Unit-6: SoC Interconnect and Bus Architectures:** Shared bus vs point-to-point interconnects; AMBA buses: AHB, APB, AXI (conceptual + RTL view); Arbitration logic; Basic NoC concepts; Interconnect integration in SoCs. (5 hours)
- **Unit-7: Processor Architecture Case Study:** Case study of one commercial/open processor: ARM or RISC-V reference core (any one); Instruction decoding; Execution stages; Pipeline organization. (5 hours)
- **Unit-8: Introduction to SoC Design Flow:** IP-based SoC design; Peripheral integration; Hardware/software co-design overview; FPGA-based prototyping. (4 hours)

Laboratory / practical / tutorial Modules

- RTL implementation of:
 - Single-cycle and pipelined processor datapath
- Cache design (direct-mapped / set-associative)
- Interrupt and DMA controller modeling
- Bus-based SoC integration:
 - CPU + memory + DMA + UART
- **Hardware/Software Co-Verification course projects:** Design and verify a small open-source SoC integrating CPU, interconnect, and peripherals. Students document synthesis, timing, and area trade-offs. Run compiled RISC-V assembly programs on the designed SoC using co-simulation.

Text Books

1. David A. Patterson & John L. Hennessy, *Computer Organization and Design: RISC-V Edition*, Morgan Kaufmann, 2021.
2. Michael J. Flynn & Wayne Luk, *Computer System Design: System-on-Chip*, Wiley, 2011.

References

1. Patterson & Hennessy, *Computer Architecture: A Quantitative Approach*, 6th Ed., Elsevier, 2019.
2. W. Wolf, *Computers as Components: Principles of Embedded Computing System Design*, Morgan Kaufmann, 2012.